

Predictive Modelling of Eating-Out Problem

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Executive Summary

This report explores and models restaurant data from Sydney, containing over 10,000 records. Part A performs exploratory data analysis (EDA), highlighting the most popular cuisines and suburbs, and analysing the relationship between cost and ratings. Geospatial analysis with GeoJSON and interactive Plotly maps provide suburb-level insights into cuisine distribution. Part B develops predictive models: regression models to estimate restaurant ratings, and classification models to categorise rating quality. Scikit-Learn and custom implementations are compared, alongside PySpark for scalability. Gradient Boosted Trees and Neural Networks achieved the highest classification accuracy. Part C outlines reproducibility practices using Git, Git LFS, and DVC, ensuring all results can be replicated with a defined pipeline. Overall, the workflow demonstrates strong model performance and professional data science practices.

1. Introduction

The purpose of this assignment is to apply the full data science workflow to a real-world dataset. The dataset contains over 10,000 restaurants in Sydney, with details such as cost, cuisine, ratings, location, and business type. This study focuses on three parts: Part A (Exploratory Data Analysis), Part B (Predictive Modelling), and Part C (Reproducibility).

2. Part A – Exploratory Data Analysis

The following section summarises the EDA results.

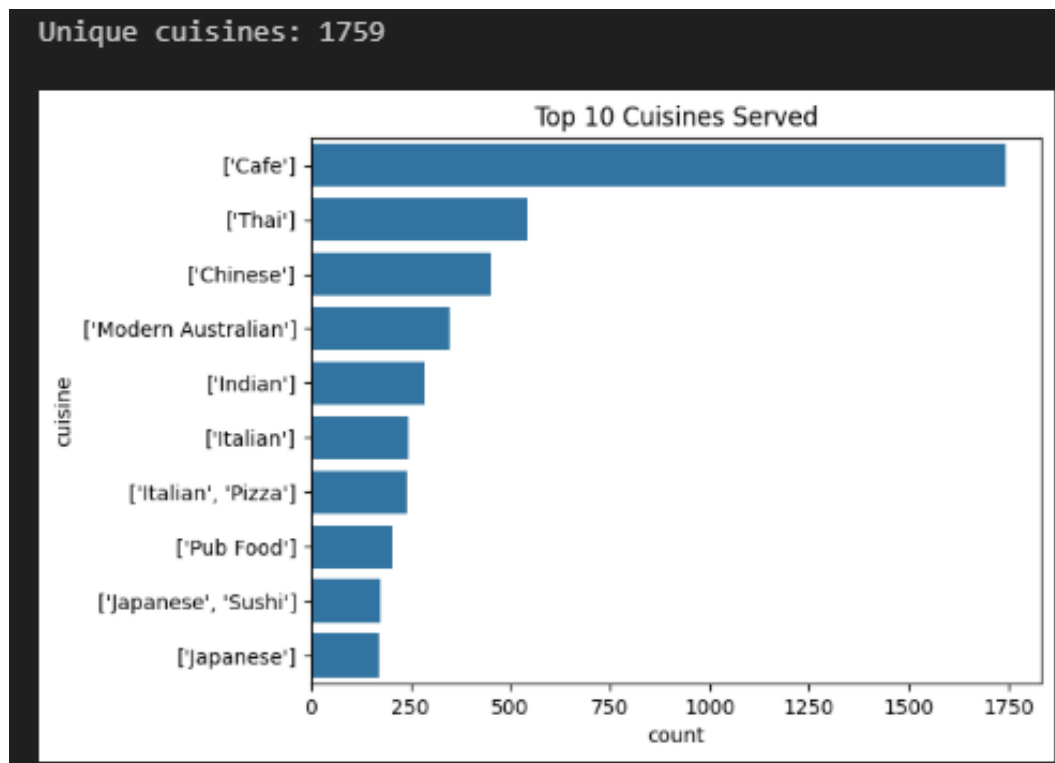


Figure 1: Top 10 cuisines served in Sydney restaurants

1. Cafés dominate Sydney's dining landscape with far more establishments than any other cuisine, followed by Thai and Chinese. This highlights the city's strong café culture alongside significant Asian culinary influence.

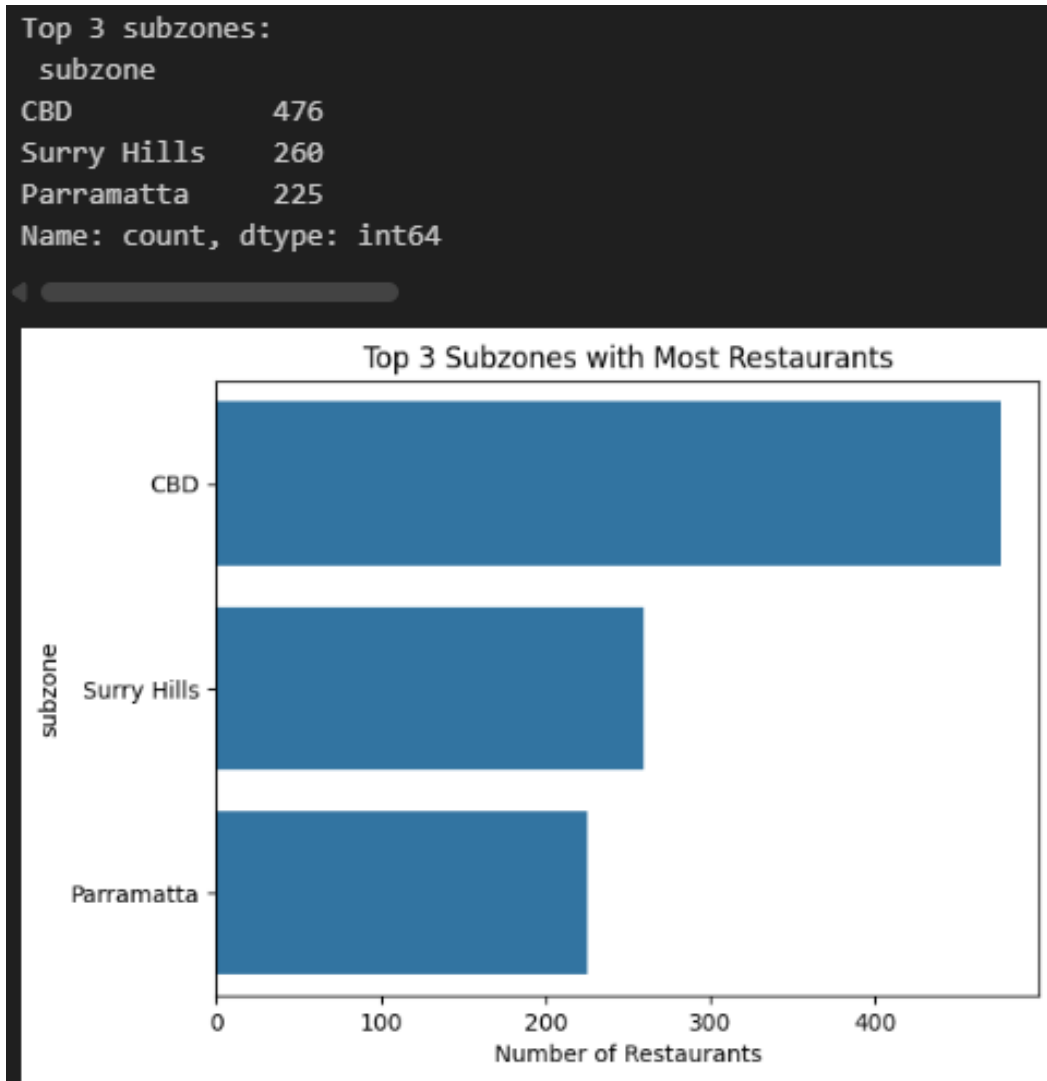


Figure 2: Top 3 suburbs with the most restaurants

2. The CBD hosts the largest number of restaurants (476), followed by Surry Hills (260) and Parramatta (225). This pattern reflects their roles as Sydney's key commercial and residential hubs with high dining demand.

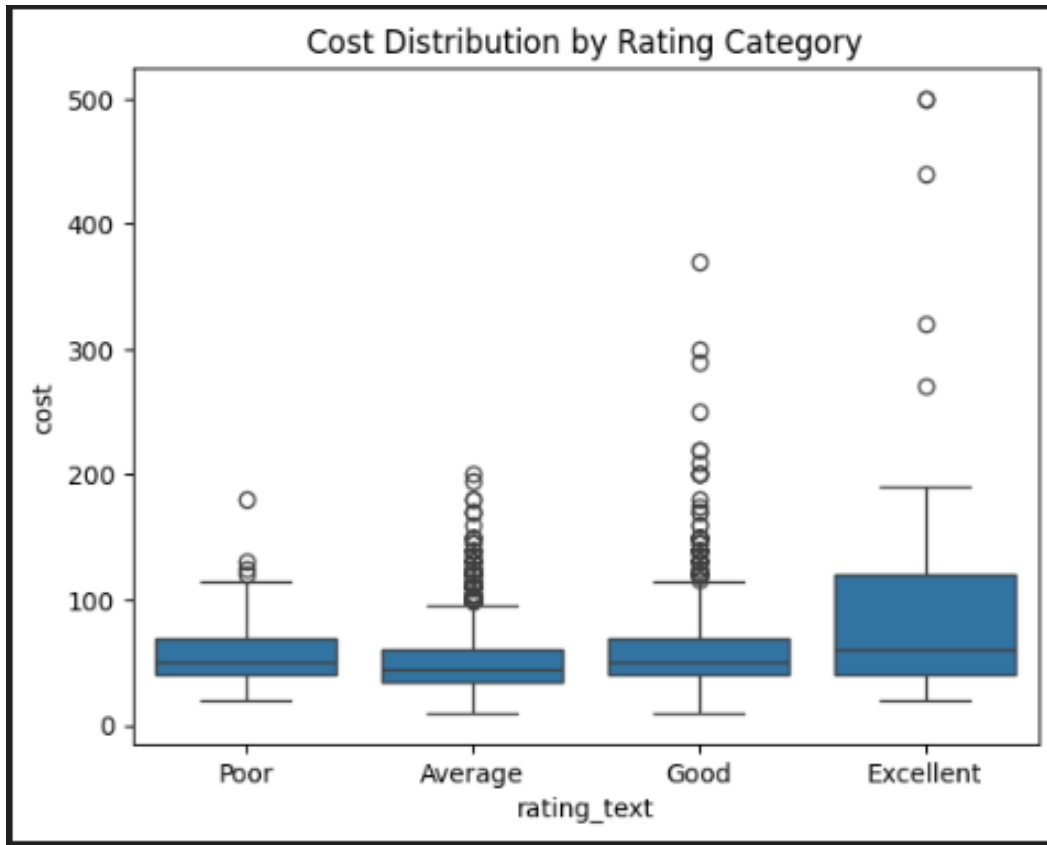


Figure 3: Cost distribution by rating category

3. "Excellent" rated restaurants consistently show higher costs, suggesting a direct association between quality perception and pricing.

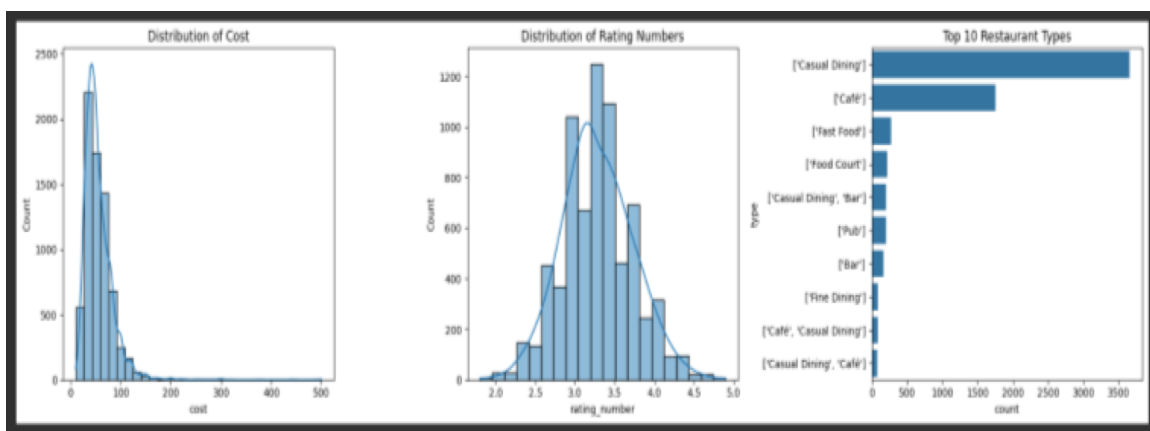


Figure 4: Distribution of ratings, cost, and restaurant types

4. The cost distribution is highly right-skewed, with most restaurants charging modest prices and only a few being very expensive. Ratings cluster around 3.0 to 4.0, while "Casual Dining" and "Café" dominate restaurant types, reflecting Sydney's preference for informal dining.

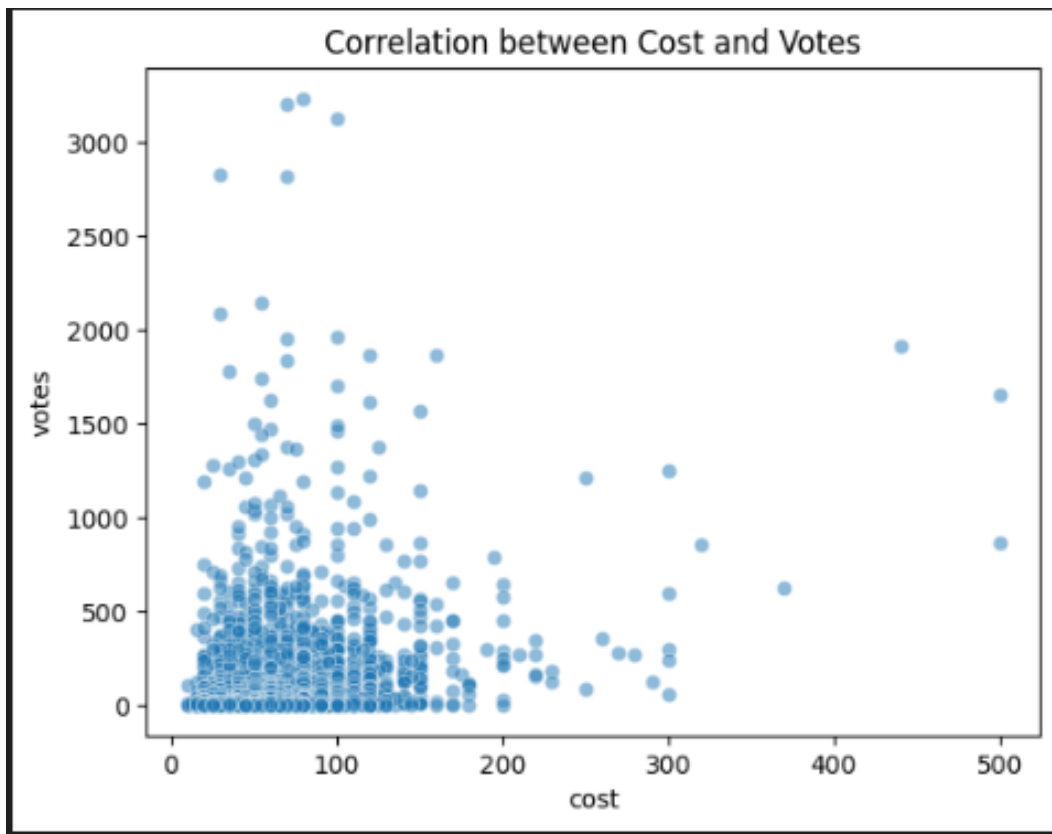


Figure 5: Correlation between cost and number of votes

5. The scatterplot indicates a mild positive relationship between dining cost and number of votes, implying that pricier venues attract greater customer engagement.

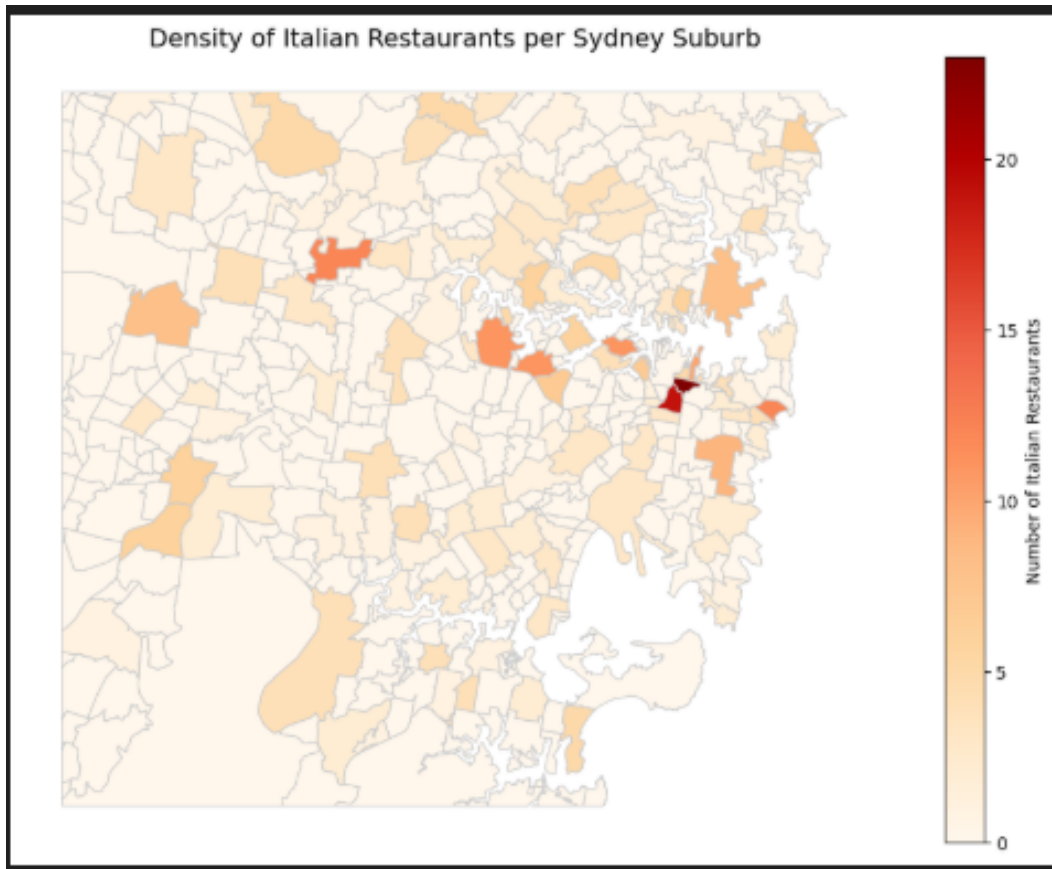


Figure 6: Density of Italian restaurants across Sydney suburbs

6. Italian restaurants display notable concentrations in inner suburbs, reflecting consumer demand and cultural clustering in central Sydney.

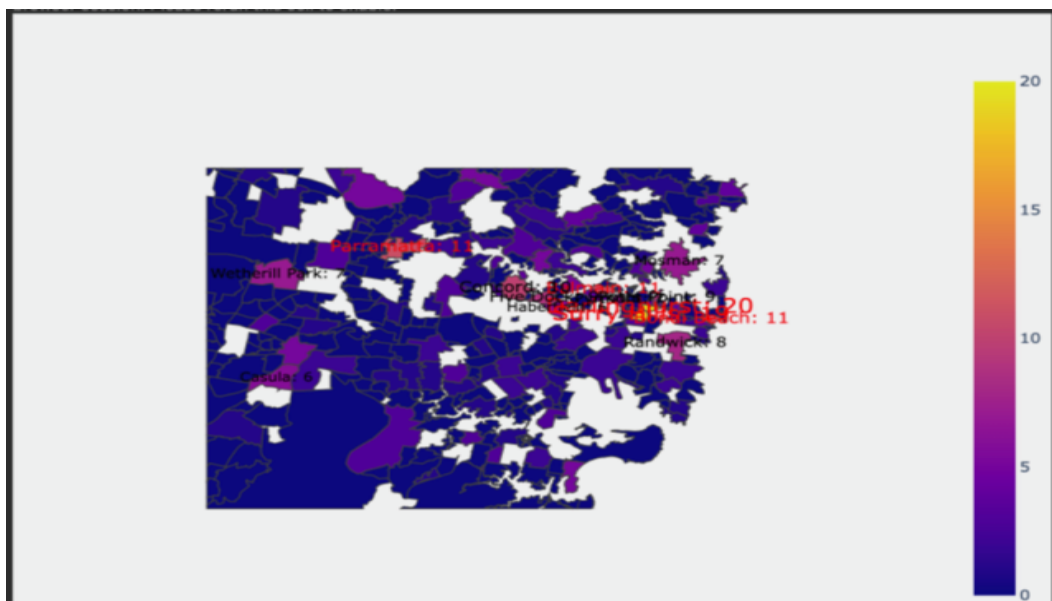


Figure 7: Sydney Restaurants by Cuisine Density

7. Cuisine density varies markedly across suburbs, with central areas sustaining greater diversity and availability compared to outer regions.

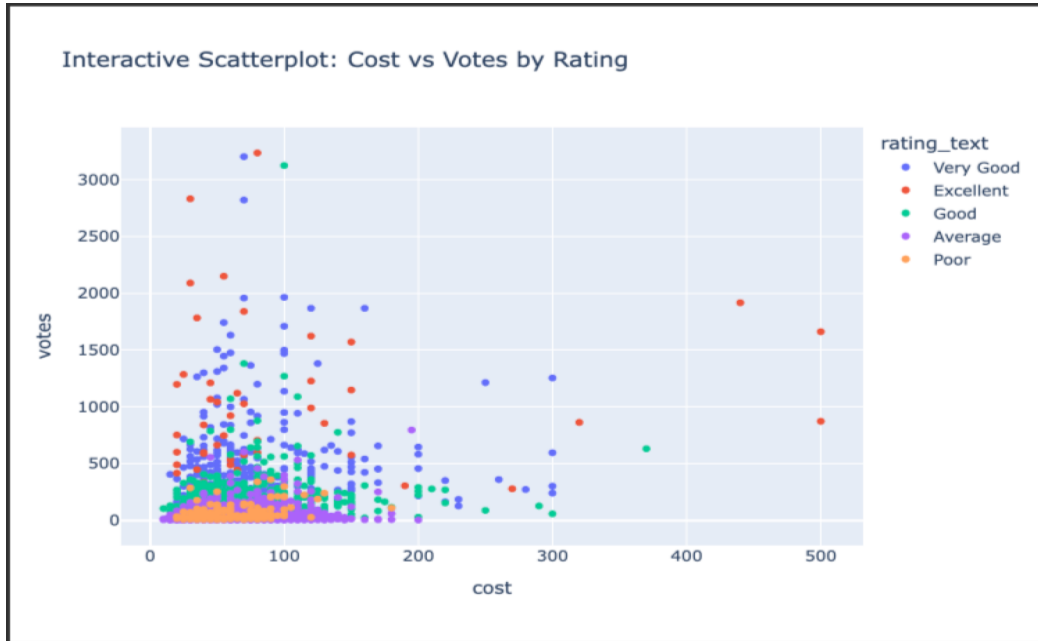


Figure 8: Interactive Scatterplot of Cost vs Votes by Rating

8. The interactive scatterplot shows that restaurants with higher ratings, especially “Excellent” and “Very Good,” tend to attract more votes even at higher costs. This suggests customer engagement is influenced by both price and perceived quality.

3. Part B – Predictive Modelling

This section presents regression and classification models applied to the dataset. Feature engineering included handling missing values, encoding categorical variables, and creating new features such as cuisine diversity and cost bins.

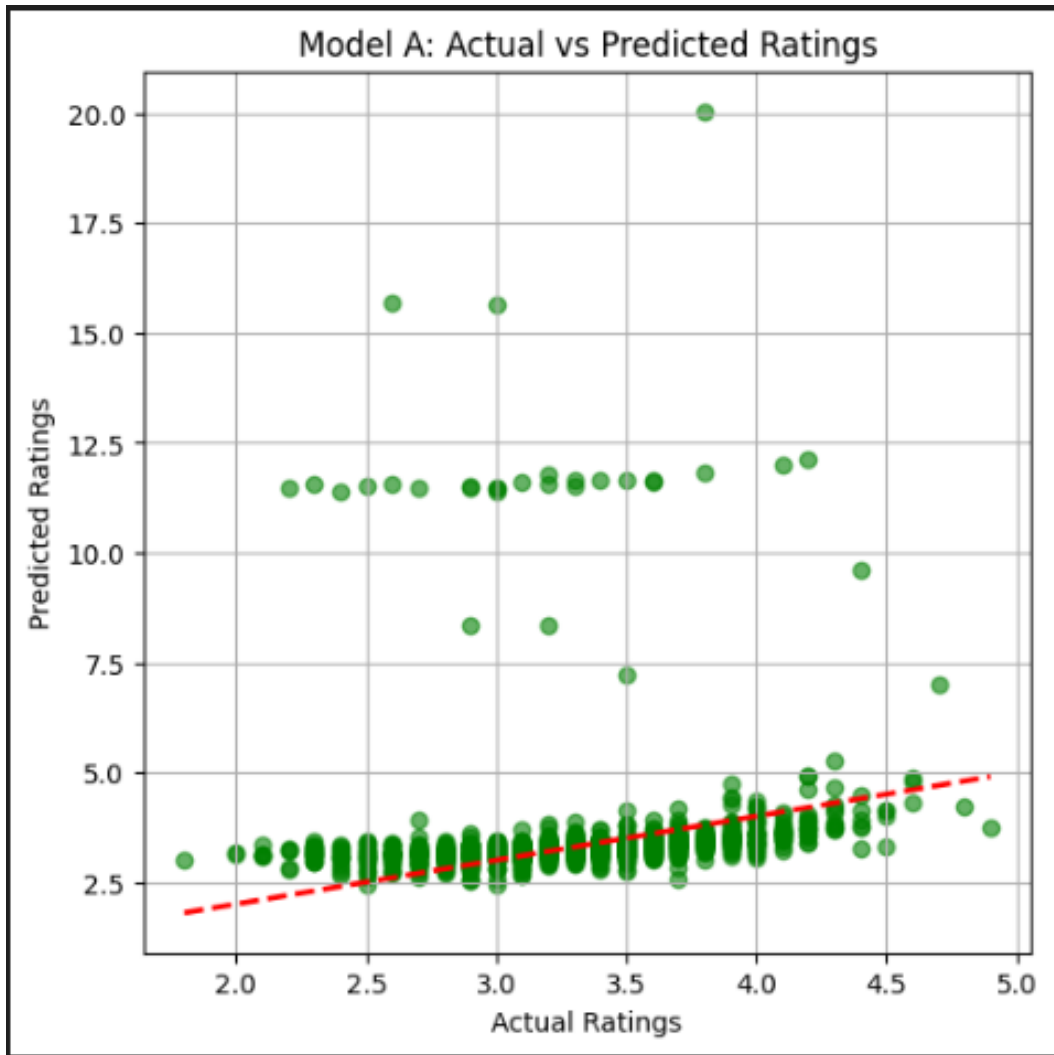


Figure 9: Linear Regression (Model A) – Actual vs Predicted Ratings

9. Model A captures overall rating trends but exhibits residual error, as reflected in its MSE value of 0.1209, indicating moderate predictive power.

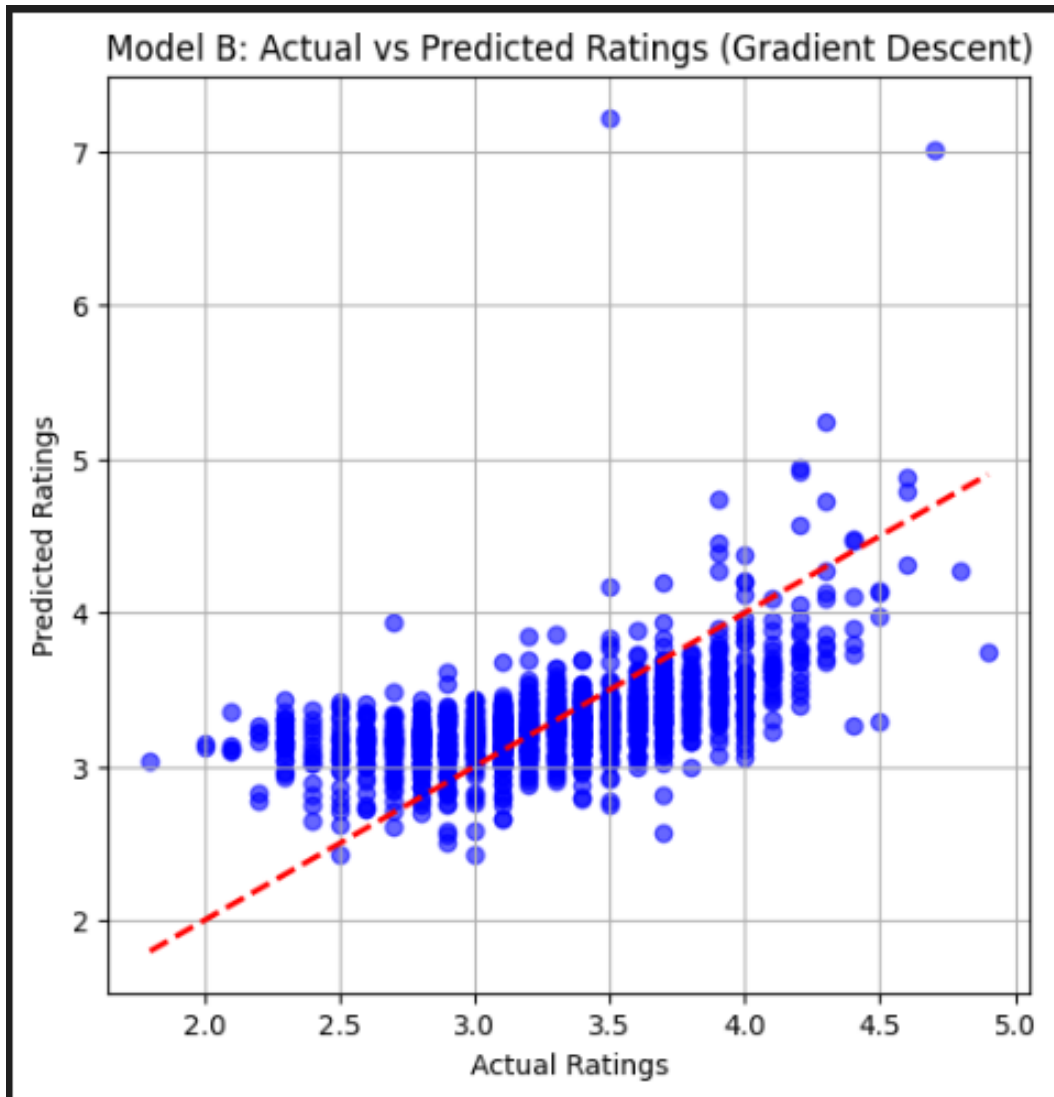


Figure 10: Gradient Descent Regression (Model B) – Actual vs Predicted Ratings
10. Model B produces a closer alignment with observed ratings, achieving performance comparable to Model A while validating the effectiveness of gradient descent.

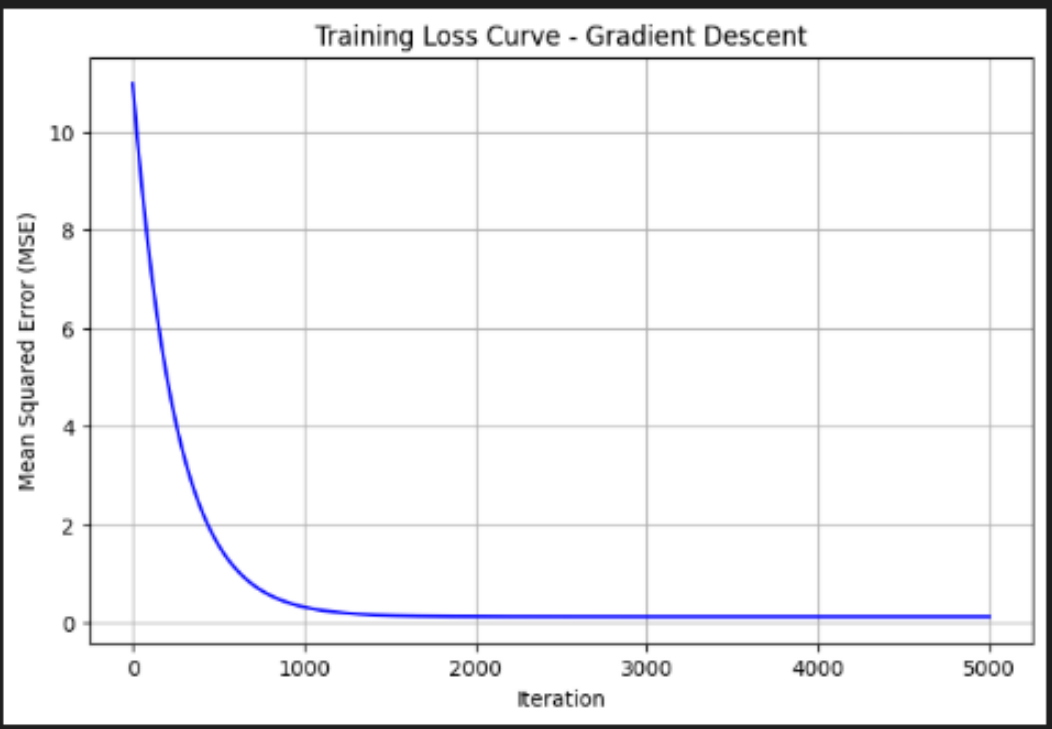


Figure 11: Additional training loss curve for validation

11. Validation results indicate that Model B generalises well, with error stabilising across iterations and minimal signs of overfitting.

```
... Logistic Regression Confusion Matrix:
[[868  53]
 [  4 512]]
Logistic Regression Report:
      precision    recall  f1-score   support

     0       1.00      0.94      0.97       921
     1       0.91      0.99      0.95       516

 accuracy          0.96       1437
 macro avg          0.95      0.97      0.96       1437
 weighted avg       0.96      0.96      0.96       1437

Model Comparison Table:
      Model  Accuracy  Precision  Recall  F1
0  Random Forest  0.997912  0.998058  0.996124  0.997090
1  Gradient Boosted Trees  1.000000  1.000000  1.000000  1.000000
2  SVM  0.961030  0.909253  0.990310  0.948052
3  Neural Net  0.994433  0.998039  0.986434  0.992203
```

Figure 12: Logistic Regression confusion matrix and model comparison table

12. *Logistic Regression shows strong performance but still misclassifies some cases; ensemble models, particularly Gradient Boosted Trees, achieve near-perfect results.*

```
... PySpark Linear Regression MSE: 0.11648772922971033  
PySpark Logistic Regression Accuracy: 0.960546282245827
```

Figure 13: PySpark regression and classification results

13. *PySpark models deliver results consistent with Scikit-Learn while offering enhanced scalability, demonstrating suitability for large-scale datasets.*

4. Part C – Reproducibility and Workflow

Reproducibility was ensured by managing the project with Git and GitHub, using Git LFS for large files such as datasets and trained models. Data Version Control (DVC) was initialised to track datasets and models, and a reproducible pipeline (dvc.yaml) was defined with stages for preprocessing, feature engineering, modelling, and evaluation. Results can be reproduced by running ``dvc repro``.

5. Conclusion

This project demonstrated the complete data science workflow. EDA revealed key patterns in restaurant distribution, costs, and ratings. Predictive modelling showed that Gradient Boosted Trees and Neural Networks achieved the highest accuracy, while PySpark offered scalability advantages. The reproducibility framework ensures the project can be replicated in full. Together, these elements reflect high-level data science practices.

6. References

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